

# Extraction and Evaluation of Caries in Dental X-ray using UNet: A Study

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**Abstract-** Monitoring of oral health is one of the prime tasks and the untreated oral health (OH) will lead to other health issues. Tooth caries is one of the common OH problem and early detection and treatment can help to prevent the caries. This research aims to develop and implement UNet-variants based scheme to extract and evaluate the caries from the chosen image. The various phases in the proposed scheme consist the following; image and mask collection and resizing it to 224x224 pixels, implementing UNet model for extracting the caries, comparing the segmented caries with mask, and computing the necessary metrics to verify the merit of the implemented technique on the chosen model. In this work, a detailed analysis on the various UNet schemes are performed to extract the necessary region from the chosen tooth X-ray and the outcome of the UNet+ helped to provide a better result on the chosen model compared to other UNet-variants and SegNet model.

**Keywords**—Healthcare; Tooth issue; Caries; Segmentation; Evaluation.

## I. INTRODUCTION

Due to its clinical significance, a number of Computer Algorithm (CA) based methods are proposed and implemented in the literature to analyse the medical images collected using a chosen imaging modality. The CA considered for a chosen task will help to automate the examination task, which considerably reduces the diagnostic burden without affecting the diagnostic accuracy. Further, the diagnostic reports are easily stored and retrieved based on the need. Along with the data analysis, this technique also supports the disease evidence information preservation task for the future analysis [1].

Dental abnormality is one of the common issues in individuals and timely diagnosis and treatment is very essential. Appropriate treatment will help to completely cure most of the dental issues. Tooth Caries (TC) is one of the common issues in individuals and untreated TC will lead to various issues, including loss of teeth. It is a common dental health concern affecting individuals of all ages. The problem results from the deterioration of the tooth's enamel due to bacterial activity [2-4].

Before initiating the treatment, it is necessary to identify the location and the severity of the TC. To detect this, a number of methods are suggested, like personal inspection by a dentist, digital camera based detection, and dental X-ray assisted analysis. The personal check and digital image will help to detect the common location of the TC and the X-ray supported approach helps to provide the necessary information

like the severity of the caries. Hence, dental X-ray based technique is commonly adopted to analyse the TC in individuals along with the personal/camera based detection procedures.

The earlier works on caries detection based on digital photograph can be found in [5] and dental X-ray based detection can be accessed from [6,7], in which the chosen X-ray image is analysed using the CA for detecting the abnormality in the teeth. The earlier works employed the segmentation method to extract and evaluate the caries region for further analysis. As discussed in [1], the collected information can be preserved in a chosen location for further analysis, if necessary.

Image supported examination of caries with Deep-Learning Tool (DLT) is commonly adopted to achieve a better accuracy [8,9]. Hence, this research also aims to develop a DLT to extract and evaluate the TC from the chosen X-ray images. The various stages in the proposed technique includes the following stages; image and mask collection form the chosen database and resizing it to 224x224 pixels, image augmentation to increase image and mask to 500-numbers, training the DLT with the chosen images, extraction of the binary caries region, and comparing the mask with extracted TC to compute the necessary image metrics.

In this work, the segmentation technique is implemented using the UNet, UNet+, UNet++, and SegNet models and the achieved results are verified. In order to improve the segmentation result, this work considered the image enhancement based on adaptive-thresholding, which improves the visibility of the caries considerably and helps to achieve a better TC detection.

This work helped to provide a segmentation accuracy of >98% with UNet, which is better compared with other methods implemented in this study. In the future, the performance of the TC segmentation can be improved by enhancing the X-ray images with a chosen thresholding procedure. Further, the performance of the proposed DLT needs to be verified using the clinically collected X-ray images, to verify the clinical significance of the developed system.

This research contributes the following;

(i) Initially executing the necessary TC detection using the UNet models,

(b) Verifying the performance of UNet and SegNet techniques and comparing the result to identify the best model for the chosen X-ray database.

Remaining section of this study consist the following regions. Section 2 provides motivation, Section 3 demonstrate the executed technique and its outcome, and Sections 4 and 5 offers results and conclusion.

## II. CONTEXT

Due to its clinical significance, a number of CA is developed to resolve a variety of image assessment tasks, like classification and segmentation. Due to its importance and better results, the DLT are widely adopted in the image examination related tasks [10,11].

The dental X-ray play a vital role in analysing a variety of tooth data and the outcome of this analysis will help to plan and execute the necessary treatment. Recently, the examination of the caries from X-ray is discussed by a number of researchers, to analyse the caries and normal region from the chosen image. During this task, the images available in a chosen repository are considered for the examination and the outcome is then considered to take the necessary decision regarding the tooth changes.

This work considered the recently discussed research by Ramya et al. (2023), which implemented DLT segmentation scheme to analyse the TC [12]. This work implemented UNet approach and achieved better segmentation accuracy during whole tooth extraction in X-ray. The considered database in the earlier work consist image and the mask to execute the DLT based segmentation.

Improving the segmentation accuracy for the chosen X-ray database is the motivation of the current research work. In order to improve the segmentation accuracy, this work implemented the adaptive-thresholding based image enhancement and the DLT segmentation to improve the result. After implementing the suggested technique, this work presented the analysis between the chosen DLT models and this outcome confirms that the UNet+ helped to achieve a detection result of >98%.

## III. METHODOLOGY

Improving detection outcomes is a primary requirement during the image analysis, and the methods provided in this study are illustrated in Figure 1.

The annotated image and the mask are first obtained from the selected database and subsequently resized to 224x224 pixels [5]. Post-resizing, the images are augmented and then enhanced using adaptive-thresholding, and these images are subsequently utilised for training the selected DLT. In this training challenge, the picture and mask are utilised to enable the DLT to differentiate between abnormal and normal areas. In the mask, caries pixels are allotted a value of '1', while other areas are allocated a value of '0'. Subsequent to the training, the efficacy of the selected DLT is evaluated, resulting in the acquisition of a binary image. This binary image is later compared with the mask to calculate the values of TP, TN, FP, and FN after which the requisite metrics depicted in Figure 1 are generated.

The performance of the created scheme is validated based on the attained mean value of these measures. Figure 1 distinctly illustrates the phases of image acquisition to validation.

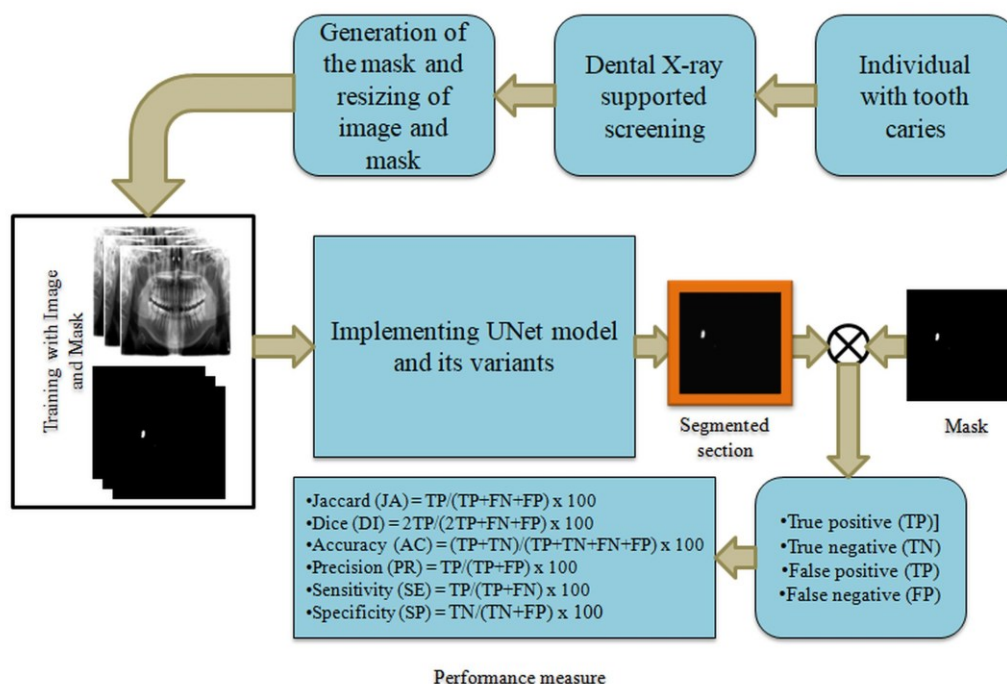


Fig. 1. Proposed scheme for Caries detection from tooth X-ray

### A. X-ray database

The practical significance of the selected technique is validated by genuine photographs, specifically utilising the dental X-ray available in [5]. Upon acquiring the tagged picture and mask, each image is scaled to 224x224 pixels and subsequently analysed.

The requisite data augmentation procedure is performed to elevate the quantities of images and masks to 500, and the data partition utilised in this study is presented in Table 1. Figure 2 displays the sample photos from the selected database. The greyscale image is augmented using adaptive thresholding, and the resultant image is subsequently analysed using the selected DLT.

TABLE I. DENTAL X-RAY IMAGE DATA SPLIT

| Class | MRI-slices |       |          |      |
|-------|------------|-------|----------|------|
|       | Total      | Train | Validate | Test |
| Image | 500        | 400   | 50       | 50   |
| Mask  | 500        | 400   | 50       | 50   |

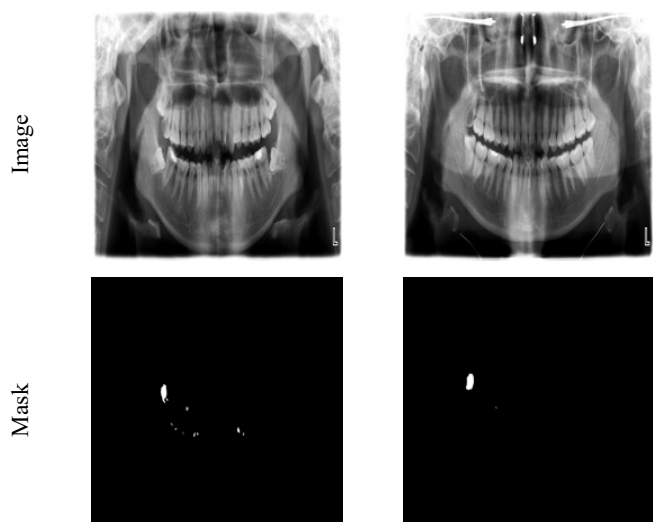


Fig. 2. Sample image and mask from the database

### B. Segmentation and Evaluation

Segmentation is a prevalent image analysis technique that facilitates the extraction of a specific region from the selected image for subsequent analysis. This study segments the carious region using DLT and subsequently assesses it through a comparative comparison with the mask. This study examined established approaches such as UNet, UNet+, UNet++, and SegNet, with pertinent information regarding these models available in [13,14]. The starting parameters for these models are designated as follows: monitoring values = accuracy and loss, batch size = 8, activation = ReLU, pooling = Max, optimisation = Adam, learning rate = 1e-4, and epochs = 100.

The proposed inquiry is conducted using Python on a computing equipment equipped with an Intel Core i5 processor, 16 GB of RAM, and a 4 GB graphics card. The

result of this procedure is the binary picture, which is subsequently compared with the mask to calculate the required metrics, as illustrated in Figure 1.

The efficacy of the DLT is substantiated by the metrics presented in Figure 1, hence validating the performance of the proposed value. This study considers the final set of 50 training photos for verification purposes. The average of the measurements is selected to validate the efficacy, confirming that the suggested technique achieved an accuracy of 98%. The practical significance of the selected technique is validated by genuine photographs, specifically utilising the dental X-ray available in [1]. Upon acquiring the tagged picture and mask, each image is scaled to 224x224 pixels and subsequently analysed.

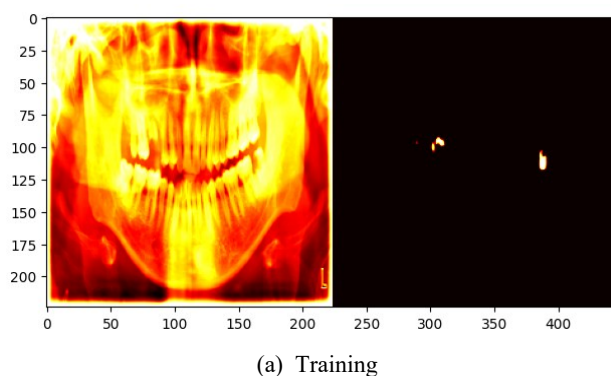
The requisite data augmentation procedure is performed to elevate the quantities of images and masks to 500, and the data partition utilised in this study is presented in Table 1. Figure 2 displays the sample photos from the selected database. The greyscale image is augmented using adaptive thresholding, and the improved image is subsequently utilised for analysis with the selected DLT.

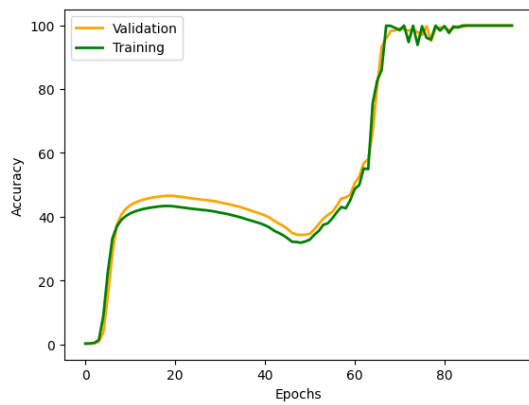
## IV. RESULT AND DISCUSSION

This section of the study delineates the experimental results and the analysis of the obtained findings. The UNet-based method is first applied to the selected image database, and the testing results are utilised for evaluation.

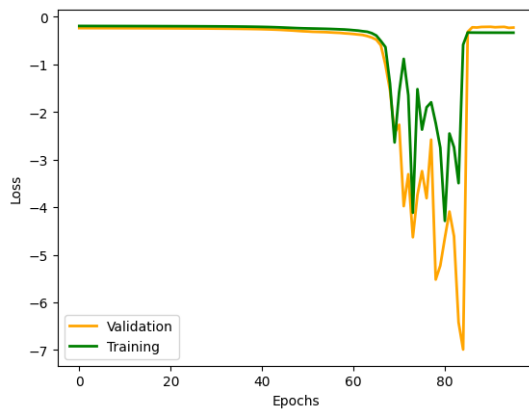
Figure 3 illustrates the training and testing outcomes obtained using the UNet model. Figure 3(a) illustrates the integration of the picture and mask (hot colour map) utilised during the training of the UNet. Figures 3(b) and (c) illustrate the accuracy and loss values attained across the selected epochs. The photos validate that the suggested technique effectively learns from the selected image database and yields superior results. Figure 3(d) displays the testing outcome, including the picture, mask, and segmented region for a selected image.

The pixel values of the mask and segmented area are essentially equivalent, so validating the efficacy of the UNet model on the selected data. Upon evaluating each selected image, the requisite metrics are calculated, and the average values of these metrics are utilised to assess the efficacy of the proposed method.

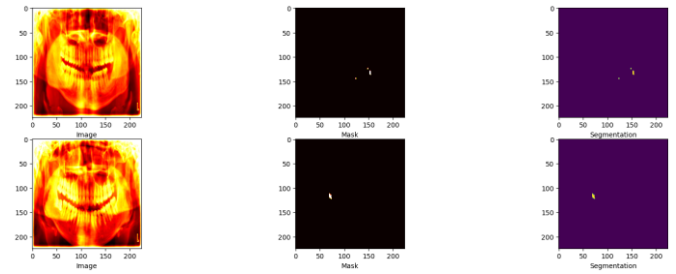




(b) Accuracy



(c) Loss



(d) Testing

Fig. 3. Results of UNet during X-ray examination

The technique is subsequently performed with additional DLTs, such as UNet+, UNet++, and SegNet, and the final results (mean) are documented in Table II for further assessment. The results affirm that the UNet+ methodology outperforms the alternative approaches selected in this study. Table II values demonstrate that the segmentation of the TC section is superior in the suggested method, achieving an accuracy exceeding 96%. This verifies that the applied approach performs effectively on the selected picture database.

Figure 4 illustrates the comparative analysis, confirming that the performance of UNet+ surpasses that of the other methods selected in this study. In the future, picture enhancement may be executed utilising a selected image thresholding task, and the presented research can be replicated with the chosen models to achieve improved segmentation accuracy.

TABLE II. OVERALL PERFORMANCE WITH CHOSEN DATABASE

| Scheme | JA    | DI    | AC    | PR    | SE    | SP    |
|--------|-------|-------|-------|-------|-------|-------|
| UNet   | 85.42 | 94.51 | 97.16 | 96.83 | 96.59 | 96.04 |
| UNet+  | 86.19 | 96.77 | 98.03 | 97.88 | 97.14 | 96.83 |
| UNet++ | 85.83 | 95.38 | 97.42 | 97.32 | 96.83 | 96.18 |
| SegNet | 84.72 | 94.86 | 97.15 | 96.79 | 96.11 | 96.32 |

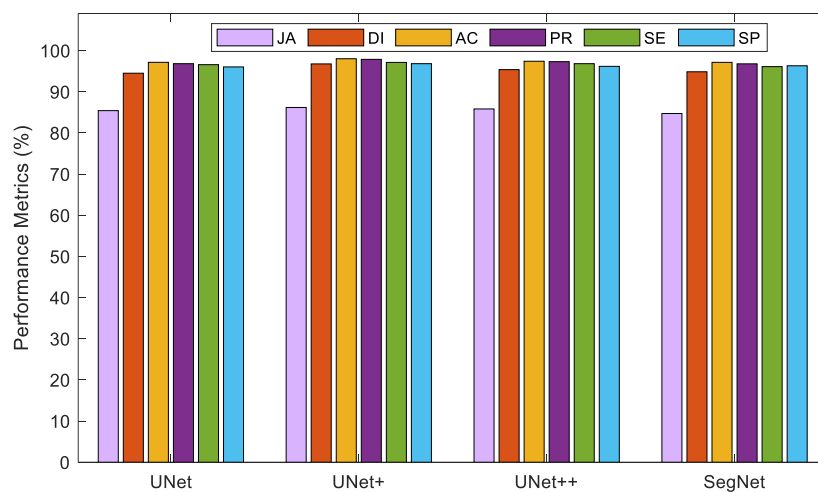


Fig. 4. Comparison of accuracy with DLT methods

## V. CONCLUSION

The analysis of digital images using a selected CA is crucial in multiple domains for enhancing the accuracy of the information derived from these images. The dental X-ray examination for TC monitoring is a crucial study area, and the results of this process will aid in the implementation of necessary actions. This study involved the analysis of caries from the selected image, employing UNet+ based segmentation to achieve improved results. The results of this procedure are compared with selected DLT methods, and based on the outcomes; the efficacy of the proposed system is validated. This study examined the adaptive-thresholding improved X-ray, and the proposed UNet+ segmentation achieved a segmentation accuracy exceeding 98%.

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